REAL LIFE 3D WAREHOUSE VISIBILITY USING Z-BUFFER

DEPTH-BASED RENDERING FOR SMARTER, CLEARER SPACES

ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

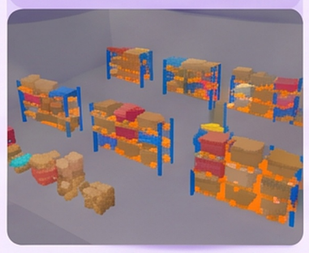
01 WHAT IS Z-BUFFER?

Z-Buffer is a depth-buffer technique used in 3D graphics to determine which objects are visible from the camera, by comparing their distance (Z-value).

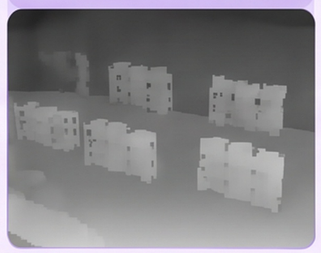
- Keeps only the closest pixels
- Hides objects behind others
- Creates a realistic 3D view

02 HOW IT WORKS?

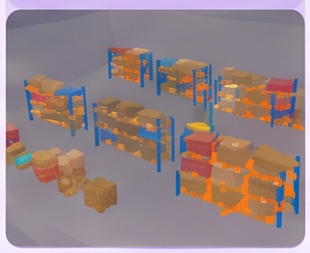
1. Render Scene



2. Build Z-Buffer



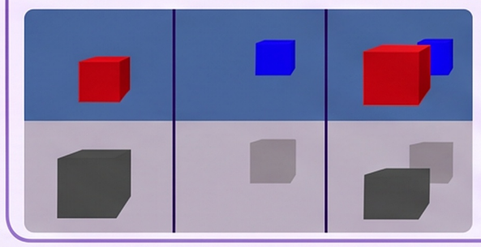
3. Compare Depth



Result: Only visible objects are rendered, giving a clear 3D view!

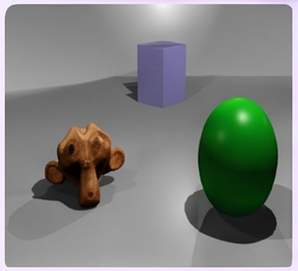
03 KEY FEATURES

- Depth-Based Visibility
- Accurate Occlusion Handling
- Better Performance
- Realistic 3D Rendering





06 DEPTH REPRESENTATION

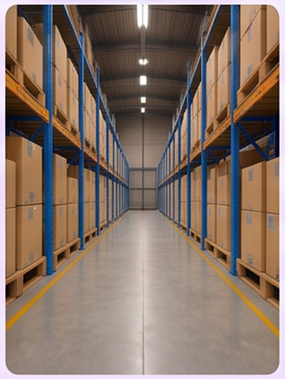


Z-Buffer shows shepth by storing distance value.



Closer objects appear brighter, farther onces ones darker.

04 REAL LIFE EXAMPLE



In a warehouse, Z-Buffer helps us see only the visible shelves and boxes, even when objects are stacked or placed behind others.

→ Enables better navigation, inventory tracking and safety!

05 APPLICATIONS

- Warehouse Management
- 3D Gaming & Simulation
- Robotics & Automation
- AR / VR Environments

07 ADVANTAGES & IMPACT

- Improves 3D realism in and industrial applications
- Helps in efficient rendering and resource usage
- Enhances user experience in real-world systems

08 CONCLUSIONS

Z-Buffer is a simple pet powerful technique that brings 3D scenes to life. It plays a vital role in modern technologies like smart warehouses, games and virtual reality.

KEY TAKEAWAY | *With Z-Buffer, we can see what matters — only the visible, in perfect depth!*

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